

Learn

FULL STACK JAVA DEVELOPMENT

Build Real Projects | Gain Practical Skills | Start Your Career

Duration: 6 Months

Mode: Online + Live Mentor Support

Software Engineering

P

PRACTICAL TRAINING
Over 120+ Hours of Live Instructor-Led Interactive Training

E

EXPERT MENTORSHIP
Direct 1-on-1 Mentorship from Senior Software Architects

C

CAREER OUTCOMES
Industry-Standard Microservices & Cloud-Native Architecture

1

Hands-on Learning

2

Real World Projects

3

Expert Guidance

4

Career Support

What's Inside This Brochure

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Explore Our Courses (14)

Turn Your Ideas Into Real Websites and Applications

About Program

Master Java, Spring Boot, Microservices, Hibernate, REST APIs, and React to build scalable enterprise web applications. Enterprise Backend & Modern Frontend Engineering

LEARNING FORMAT	DURATION & PACE	DELIVERY MODE	CREDENTIALS
Mentor-led Live	6 Months	Online + Live Mentor Support	Industry Certified
Interactive Sessions	Daily Practice Drills	Hands-on Code Labs	Marvel Slice Verified

PILLAR 01	PILLAR 02	PILLAR 03
Scalable Microservices	Applied Production Drills	Career & Placement Velocity
Architect high-throughput REST APIs, database schemas, ACID transactional models, and secure JWT auth.	Implement caching layers, message queues, containerized microservices, and automated testing suites.	Deploy full-stack cloud backends with verified Git commits, engineering code reviews, and mock interviews.

What You Will Achieve

v Design and implement secure, high-throughput RESTful and microservice backend architectures	v Model relational and NoSQL databases with ACID compliance, indexing, and query optimization
v Implement robust authentication, authorization (RBAC/JWT), and data protection mechanisms	v Write comprehensive unit, integration, and end-to-end tests following TDD methodologies
v Containerize applications with Docker and configure continuous delivery (CI/CD) pipelines	v Profile memory, monitor production logs, and optimize server response latency
v Collaborate using enterprise GitFlow branching models and automated build verification	v Ship production-ready APIs and backend services backed by comprehensive API documentation

Key Highlights

- ✓ 620+ Hrs of Applied Practical Learning
- ✓ 218+ Hrs of Self-Paced Video Modules
- ✓ 32+ Industry Projects & Real Case Studies
- ✓ 100% Dedicated Placement Assistance Support
- ✓ 24*7 Mentor Doubt Clearing Support
- ✓ 1:1 Technical & HR Mock Interviews
- ✓ 2 Days Campus Immersion & Hackathons
- ✓ Certification in Cloud & Microservices
- ✓ Up to Rs. 50 Lakhs Startup Incubation Support*
- ✓ Flexible Weekday & Weekend Batches
- ✓ 90+ Live Instructor-Led Faculty Sessions
- ✓ Learn from Senior Industry Practitioners
- ✓ Regular One-on-One Technical Mentorship
- ✓ Resume Preparation & LinkedIn Optimization
- ✓ Designed for Working Pros & Freshers
- ✓ No-Cost EMI Payment Options Available
- ✓ Top Batch Performers Fellowship Awards*
- ✓ 3 Guaranteed Placement Interviews
- ✓ Lifelong Engineering Alumni Network
- ✓ Continuous Cloud Sandboxes & LMS Access

Program Pedagogy



Instructor-led Training

Get trained by top industry experts



Self-paced videos

Learn at your own pace with world-class content



Hackathons

Get a sense of how real projects are built



Gamified Learning

Get involved in group activities to solve real-world problems



Dedicated Learning Management Team

To help you with your learning needs



Projects and Exercises

Get real-world experience through projects



Peer Networking and Group Learning

Improve your professional network and learn from peers



1:1 Personalized Learning

Hands-on exercises, project work, quizzes, and project builds

Who Can Apply?

Open to passionate learners and professionals across diverse academic, analytical, and technical backgrounds.

- Freshers & College Graduates**

Aspiring engineers seeking practical production skills to secure their first high-growth software role.
- Working Professionals Switching Roles**

Professionals in non-tech, QA or legacy roles transitioning into modern full-stack engineering careers.
- Junior Developers & Upskillers**

Developers wanting to master modern frameworks, system architecture, and enterprise best practices.
- Career Restarters & Returners**

Individuals returning to tech after a sabbatical, refreshing with 1:1 mentor support and live builds.
- Freelancers & Indie Builders**

Builders looking to deliver end-to-end client applications with production-grade speed and reliability.
- Passionate Self-Learners**

Anyone determined to complete 8 intensive modules with daily mentor-verified coding practice.

ELIGIBILITY & TECHNICAL PREREQUISITES		
Academic Eligibility Any degree / diploma (BE, BTech, BSc, BCA, BCom, or equivalent stream).	Prior Coding Background Basic programming logic helps (covered in Module 1)	System & Hardware Setup Laptop/PC with 8GB RAM, modern web browser, and broadband internet.

Application & Onboarding Journey

- 01 SUBMIT APPLICATION & PROFILE REVIEW**

Submit your academic details and career aspirations through our admissions form for instant evaluation.
- 02 1:1 CAREER COUNSELLING & ROADMAP**

Connect with an engineering career advisor to review syllabus fit, batch timings, and learning milestones.
- 03 ENROLLMENT & LMS ONBOARDING**

Confirm your batch seat, receive your Marvel Slice student portal login credentials, and join the cohort community.
- 04 ORIENTATION & MENTOR MATCH**

Attend the live orientation session, configure your developer tooling, and start Module 1 with dedicated mentor support.

Learning Path

- 1** **Enroll & Orientation**
Complete admission, get LMS access and meet your mentor.
- 2** **Foundations First**
Start from zero - setup, basics and first hands-on labs.
- 3** **Core Concepts**
The heart of the course with daily practice and exercises.
- 4** **Tools of the Trade**
Master the professional tools used in real jobs.
- 5** **Mentor Check-ins**
1:1 doubt-clearing so you never stay stuck.
- 6** **Guided Mini Projects**
Mentor-reviewed builds that lock in every stage.
- 7** **Advanced Topics**
Deeper concepts taught with real scenarios and datasets.
- 8** **Project**
End-to-end portfolio-grade build, deployed live.
- 9** **Deployment & Portfolio**
Ship your work publicly with clean documentation.
- 10** **Resume & LinkedIn**
Profiles rewritten around the projects you shipped.
- 11** **Mock Interviews**
Practice rounds with the most-asked questions.

Certification & Referrals

Earn your certificate and get referred to hiring partners.

Program Curriculum

Comprehensive Practical Syllabus • Part 1 of 4

MODULE 1

Core Java Fundamentals & OOP

Gain in-depth practical expertise in core java fundamentals & oop through guided technical walkthroughs, exercises, and real-world implementation.

- Java Syntax & Variables
- OOP Concepts & Encapsulation
- Polymorphism & Inheritance
- Exception Handling
- Collections Framework
- Generics & Lambdas

TOOLS & ENVIRONMENT

VS Code

Git & GitHub

Chrome DevTools

Postman

KEY COMPETENCY GAINED

Independently architect and execute production-ready core java fundamentals & oop implementations.

TECHNICAL DRILL & PRACTICAL LAB

Work through step-by-step practical coding exercises and technical walkthroughs applying core java fundamentals & oop with mentor guidance.

PRACTICE FOCUS: Hands-on coding exercises, syntax drills & mentor validation.

MODULE 2

Advanced Java & Multithreading

Gain in-depth practical expertise in advanced java & multithreading through guided technical walkthroughs, exercises, and real-world implementation.

- Concurrency & Threads
- Executor Service
- File I/O Streams
- JDBC Database Connectivity
- Design Patterns in Java

TOOLS & ENVIRONMENT

VS Code

Git & GitHub

Chrome DevTools

Postman

KEY COMPETENCY GAINED

Independently architect and execute production-ready advanced java & multithreading implementations.

TECHNICAL DRILL & PRACTICAL LAB

Work through step-by-step practical coding exercises and technical walkthroughs applying advanced java & multithreading with mentor guidance.

PRACTICE FOCUS: Hands-on coding exercises, syntax drills & mentor validation.

Program Curriculum

Comprehensive Practical Syllabus • Part 2 of 4

MODULE 3

Database Design with MySQL

Apply human-centered design principles, interactive wireframing, high-fidelity prototypes, and cohesive design systems.

- Relational Schema Design
- Complex SQL Queries & Joins
- Indexing & Query Optimization
- Transactions & ACID properties
- Stored Procedures & Triggers

TOOLS & ENVIRONMENT

Figma

FigJam

Auto Layout

Miro

KEY COMPETENCY GAINED

Produce interactive high-fidelity design prototypes ready for developer handoff.

TECHNICAL DRILL & PRACTICAL LAB

Practice wireframing, interactive vector layout creation, and component auto-layout constraints in Figma.

PRACTICE FOCUS: Hands-on coding exercises, syntax drills & mentor validation.

MODULE 4

Spring Framework Core & Spring Boot

Gain in-depth practical expertise in spring framework core & spring boot through guided technical walkthroughs, exercises, and real-world implementation.

- Inversion of Control & DI
- Spring Boot Autoconfiguration
- Spring Bean Lifecycle
- Component Scanning & Stereotypes
- Application Properties & Profiles

TOOLS & ENVIRONMENT

VS Code

Git & GitHub

Chrome DevTools

Postman

KEY COMPETENCY GAINED

Independently architect and execute production-ready spring framework core & spring boot implementations.

TECHNICAL DRILL & PRACTICAL LAB

Work through step-by-step practical coding exercises and technical walkthroughs applying spring framework core & spring boot with mentor guidance.

PRACTICE FOCUS: Hands-on coding exercises, syntax drills & mentor validation.

Program Curriculum

Comprehensive Practical Syllabus • Part 3 of 4

MODULE 5

RESTful Microservices & Spring Cloud

Engineer robust backend services, secure RESTful API endpoints, database models, and transactional data pipelines.

- REST API Architecture
- Eureka Service Registry
- API Gateway & Routing
- Feign Client & Circuit Breakers
- Centralized Configuration

TOOLS & ENVIRONMENT

VS Code

Git & GitHub

Chrome DevTools

Postman

KEY COMPETENCY GAINED

Independently architect and execute production-ready restful microservices & spring cloud implementations.

TECHNICAL DRILL & PRACTICAL LAB

Work through step-by-step practical coding exercises and technical walkthroughs applying restful microservices & spring cloud with mentor guidance.

PRACTICE FOCUS: Hands-on coding exercises, syntax drills & mentor validation.

MODULE 6

Hibernate ORM & Spring Data JPA

Construct intelligent data pipelines, statistical model training, evaluation metrics, and inference integrations.

- Entity Mapping & Relations
- Spring Data Repositories
- JPQL & Criteria API
- Pagination & Sorting
- Second-level Caching

TOOLS & ENVIRONMENT

Jupyter Notebook

Pandas & NumPy

Scikit-Learn

KEY COMPETENCY GAINED

Independently architect and execute production-ready hibernate orm & spring data jpa implementations.

TECHNICAL DRILL & PRACTICAL LAB

Work through step-by-step practical coding exercises and technical walkthroughs applying hibernate orm & spring data jpa with mentor guidance.

PRACTICE FOCUS: Hands-on coding exercises, syntax drills & mentor validation.

Program Curriculum

Comprehensive Practical Syllabus • Part 4 of 4

MODULE 7

Frontend Foundations with React & TypeScript

Develop scalable single-page applications with declarative component hierarchies, dynamic state hooks, and routing.

- Modern React Components
- Hooks & State Management
- TypeScript Interfaces
- Tailwind CSS UI Design
- Axios HTTP Integration

TOOLS & ENVIRONMENT

React 19

React DevTools

Vite

React Router

KEY COMPETENCY GAINED

Develop modular component systems with state synchronization and route navigation.

TECHNICAL DRILL & PRACTICAL LAB

Build modular reusable components with state management drills, custom hook abstractions, and routing handlers.

PRACTICE FOCUS: Hands-on coding exercises, syntax drills & mentor validation.

MODULE 8

DevOps, Docker & Cloud Deployment

Configure live production environments, SSL certificates, automated version control, and multi-stage deployment workflows.

- Dockerizing Spring Boot Services
- Docker Compose Multi-container
- CI/CD Pipelines with GitHub Actions
- AWS EC2 Deployment
- Monitoring & Logs

TOOLS & ENVIRONMENT

Git & GitHub

cPanel / FTP

Vercel / Netlify

SSL Tools

KEY COMPETENCY GAINED

Deploy and configure live production websites with custom domains and SSL encryption.

TECHNICAL DRILL & PRACTICAL LAB

Practice Git branch workflows, pull-request merge resolutions, and continuous live deployment hosting.

PRACTICE FOCUS: Hands-on coding exercises, syntax drills & mentor validation.

🔍 Technologies & Tools You Will Master

- 🔍 VS Code
- 🔍 Git & GitHub
- 🔍 Postman
- 🔍 MySQL
- 🔍 Docker Basics
- 🔍 Linux Basics
- 🔍 React.js 19
- 🔍 Next.js App Router
- 🔍 TypeScript
- 🔍 Tailwind CSS
- 🔍 Redux Toolkit
- 🔍 HTML5 / CSS3 / ESNext

Course Projects

Industry-grade portfolio applications built step-by-step from beginner to advanced.

BEGINNER01

Project 1: Enterprise Multi-Vendor Marketplace & E-Commerce Platform

A comprehensive e-commerce platform supporting multiple seller storefronts, customer carts, and automated order fulfillment. Engineered with a decoupled frontend and scalable microservices

STACK: React, Node.js, Express, PostgreSQL, Redis, Stripe API,

INTERMEDIATE02

Project 2: Real-Time Collaborative Workspace & Communications Hub

A high-performance team collaboration platform featuring real-time simultaneous document editing, operational transformation, multimedia uploads, and low-latency group audio/chat channels.

STACK: TypeScript, Next.js, WebSockets, Socket.io, MongoDB, AWS

ADVANCED03

Project 3: Cloud-Native Microservices SaaS with Automated DevOps CI/CD

A subscription-based SaaS application broken down into independent microservices: User Authentication, Subscription Billing, Email Dispatcher, and Analytics Ingestion.

STACK: Node.js, Python, RabbitMQ, Docker, AWS EC2/RDS, GitHub

ADVANCED04

Project 4: AI-Enhanced Business Intelligence & Analytics Dashboard

An executive business dashboard that ingests raw operational metrics, renders interactive charts, and leverages LLM AI APIs to automatically generate executive analytical summaries.

STACK: React, FastAPI, Gemini / OpenRouter AI API, PostgreSQL,

Career Roles & Opportunities

 Backend Developer

 Full Stack Developer

 API / Integration Engineer

 Freelance Developer

Career Services & Placement Assistance

 **Career Roadmaps & Strategy**
1:1 technical career planning aligned to your profile.

 **Resume & LinkedIn Makeover**
ATS-compliant resume & recruiter-ready profile.

 **1:1 Mentor Guidance**
Personal doubt-clearing & transition counseling.

 **Technical Mock Interviews**
Practice with FAANG & enterprise interview drills.

 **Direct Hiring Referrals**
Priority shortlisting across 150+ verified hiring partners.

 **Hackathons & Hiring Days**
Live collaborative team builds to demonstrate skills.

Contact Us & Admissions Support

ADMISSIONS HELPLINE
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+91 80882 18609 - Mon-Sat 9AM-7PM

STUDENT SUPPORT & INQUIRIES
sales@marvelslice.com
Prompt admissions response within 24 hrs

CAMPUS & LEARNING CENTER
Marvel Slice - Institute for Software Learning
Chennai, Tamil Nadu, India

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OUR CORE PHILOSOPHY & PROMISE

"Education should create opportunities; skills should create confidence."

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Explore Our Courses

One brochure per course — contact our academic counsellors to receive the comprehensive curriculum for any program.

01

Full Stack Java Development
Enterprise Backend & Microservices
Mentor-Led • Live Sprints • Certified

CURRENT

02

Full Stack Web Development - MERN
React, Node, Express & MongoDB
Mentor-Led • Live Sprints • Certified

03

Python Full Stack Development
Django, FastAPI, SQL & Cloud APIs
Mentor-Led • Live Sprints • Certified

04

AI & Machine Learning (AIML)
Supervised, Unsupervised & Deep Neural Nets
Mentor-Led • Live Sprints • Certified

05

Data Science & Machine Learning
Statistical Analytics, NumPy & Pandas
Mentor-Led • Live Sprints • Certified

06

Data Analytics & Business Intelligence
Power BI, Advanced SQL, Excel & Tableau
Mentor-Led • Live Sprints • Certified

07

Generative AI & Prompt Engineering
LLMs, RAG Architectures & LangChain
Mentor-Led • Live Sprints • Certified

08

GenAI, Agentic AI & Robotics
Autonomous AI Agents & Embedded Systems
Mentor-Led • Live Sprints • Certified

09

DevOps & Cloud Engineering
Docker, Kubernetes, CI/CD & AWS
Mentor-Led • Live Sprints • Certified

10

Software Testing - Automation
Selenium, Playwright, TestNG & CI/CD
Mentor-Led • Live Sprints • Certified

11

Software Testing - Manual QA
SDLC, STLC, Agile Sprints & Bug Tracking
Mentor-Led • Live Sprints • Certified

12

UI/UX Design & Product Experience
Figma, Wireframing, Prototyping & Design Systems
Mentor-Led • Live Sprints • Certified

13

Robotics & Embedded Systems
Arduino, Raspberry Pi, Sensors & C++
Mentor-Led • Live Sprints • Certified

14

Advanced Prompt Engineering
Chain-of-Thought, Tool Calling & Fine-Tuning
Mentor-Led • Live Sprints • Certified

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